A Complete Guide to LSTM

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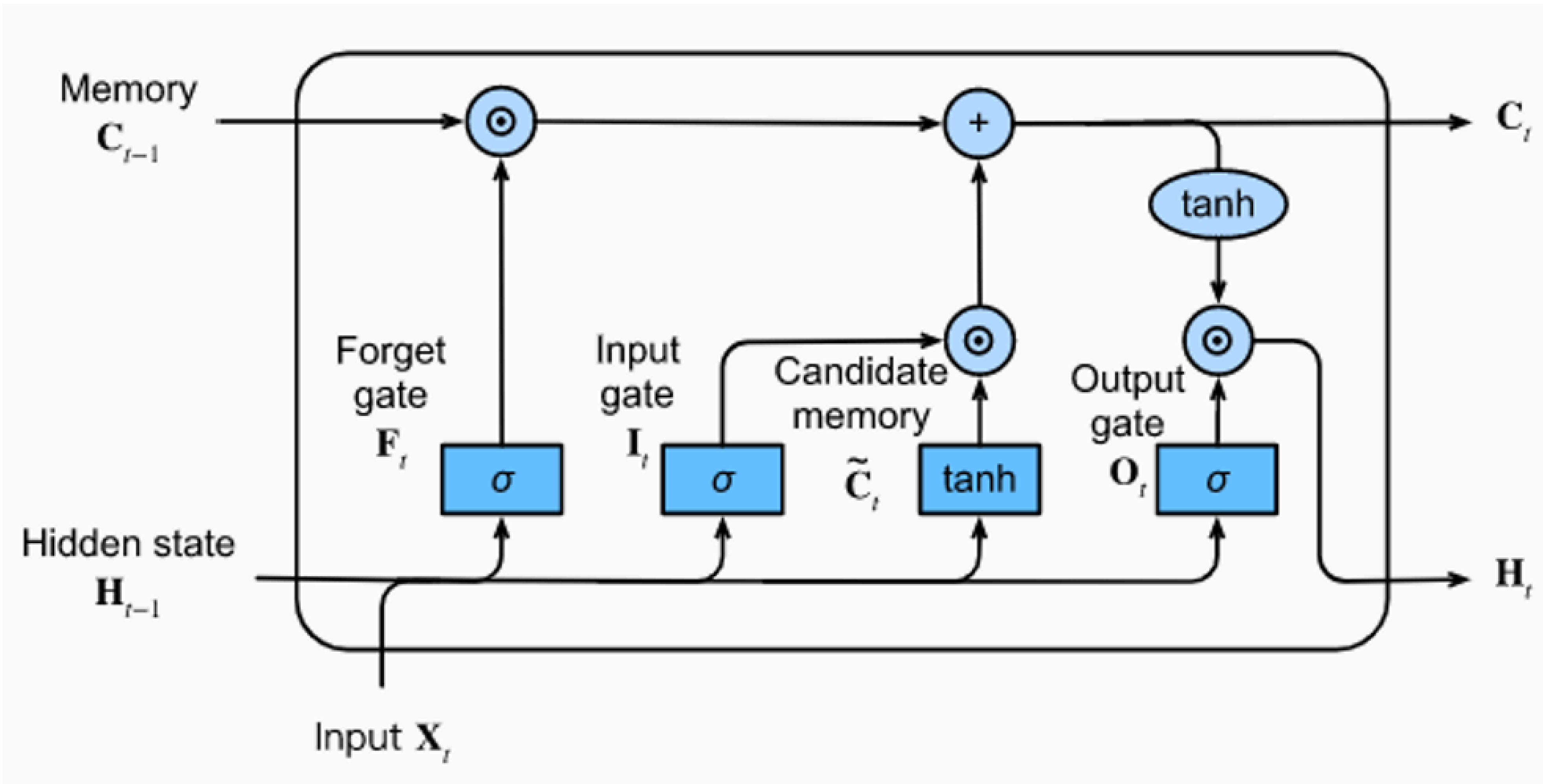
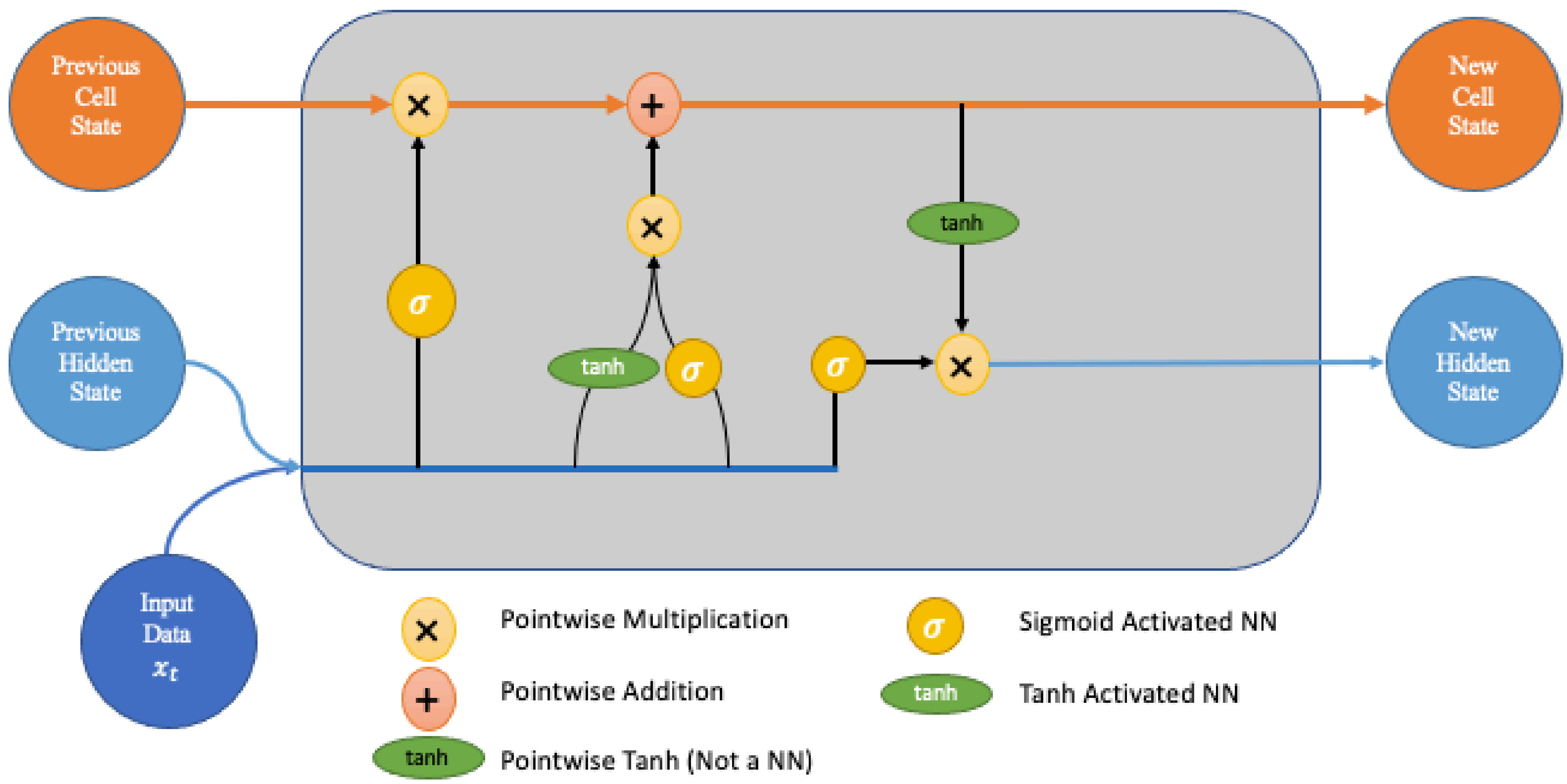
Complete Notes





1. Why LSTM Exists: The Core Problem

Before LSTM, the main model for sequence data was the **Recurrent Neural Network (RNN)**.

RNN works like this:

$$h_t = \tanh(W_h h_{t-1} + W_x x_t + b)$$

Where:

- x_t = input at time t
- h_t = hidden state
- W_h, W_x = weights

Problem: Vanishing and Exploding Gradient

During training, RNN uses **Backpropagation Through Time (BPTT)**.

Gradients get multiplied repeatedly:

$$\frac{\partial L}{\partial h_0} = \frac{\partial L}{\partial h_t} \prod_{k=1}^{t} \frac{\partial h_k}{\partial h_{k-1}}$$

If values < 1 → gradients shrink → **vanishing gradient**

If values > 1 → gradients explode → **exploding gradient**

Result:

Normal RNN cannot remember long-term information.

Example:

Sentence:

"I grew up in France... I speak fluent ____"

RNN forgets "France"

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2. LSTM: Core Idea

Introduced by:

Hochreiter & Schmidhuber (1997)

Goal:

Allow network to:

- Remember important information
- Forget irrelevant information
- Control memory flow

Key innovation:

Cell State

3. LSTM Architecture Overview

Each LSTM cell contains:

- Cell State
- Hidden State
- Forget Gate
- Input Gate
- Candidate Gate
- Output Gate

Two outputs:

$$h_t = hidden\ state$$

$$C_t = cell\ state$$

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4. Cell State — The Memory Highway

Cell state acts like a **conveyor belt**

$$C_{t-1} \rightarrow C_t$$

Information flows with minimal changes.

This prevents gradient from vanishing.

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5. Gates in LSTM

Gates use:

Sigmoid activation:

$$\sigma(x) = \frac{1}{1 + e^{-x}}$$

Output range:

0 to 1

Meaning:

0 = block completely

1 = allow completely

6. Forget Gate

Purpose

Decides:

What information to forget

Formula:

$$f_t = \sigma(W_f[h_{t-1}, x_t] + b_f)$$

Output:

Vector between 0 and 1

Update memory:

$$C_{t-1} * f_t$$

If:

0 → forget
1 → keep

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7. Input Gate

Purpose:

Decides what new information to store

Two steps:

Step 1: Gate layer

$$i_t = \sigma(W_i[h_{t-1}, x_t] + b_i)$$

Step 2: Candidate memory

$$\tilde{C}_t = \tanh(W_c[h_{t-1}, x_t] + b_c)$$

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8. Update Cell State

Combine forget and input:

$$C_t = f_t * C_{t-1} + i_t * \tilde{C}_t$$

Meaning:

Forget old info
Add new info

9. Output Gate

Purpose:

Decide what to output

Step 1:

$$o_t = \sigma(W_o[h_{t-1}, x_t] + b_o)$$

Step 2:

Hidden state:

$$h_t = o_t * \tanh(C_t)$$

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10. Final Complete LSTM Equations

Forget gate:

$$f_t = \sigma(W_f[h_{t-1}, x_t] + b_f)$$

Input gate:

$$i_t = \sigma(W_i[h_{t-1}, x_t] + b_i)$$

Candidate:

$$\tilde{C}_t = \tanh(W_c[h_{t-1}, x_t] + b_c)$$

Cell state:

$$C_t = f_t C_{t-1} + i_t \tilde{C}_t$$

Output gate:

$$o_t = \sigma(W_o[h_{t-1}, x_t] + b_o)$$

Hidden state:

$$h_t = o_t \tanh(C_t)$$

11. Dimensions

If:

Input size = d
Hidden size = h

Then:

Weights:

$$W_f = (h, h + d)$$

$$W_i = (h, h + d)$$

$$W_o = (h, h + d)$$

$$W_c = (h, h + d)$$

Bias:

$$(h, 1)$$

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12. Why LSTM Solves Vanishing Gradient

Because:

Cell state update:

$$C_t = f_t C_{t-1} + \ldots$$

Gradient flows directly.

Derivative:

$$\frac{\partial C_t}{\partial C_{t-1}} = f_t$$

If:

$$f_t \approx 1$$

Gradient preserved.

13. Forward Pass Algorithm

For each time step:

Step 1:

Forget gate

Step 2:

Input gate

Step 3:

Candidate memory

Step 4:

Update cell state

Step 5:

Output gate

Step 6:

Compute hidden state

14. Backpropagation Through Time

Training:

Loss computed:

$$L = \sum_t L_t$$

Gradient flows:

Through

- Hidden state
- Cell state
- Gates

LSTM reduces vanishing gradient significantly.

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15. Types of LSTM

A. Vanilla LSTM

Standard structure

Most common

B. Stacked LSTM

Multiple layers:

$$LSTM_1 \to LSTM_2 \to LSTM_3$$

Used in:

Deep learning tasks

C. Bidirectional LSTM

Two directions:

Forward

Backward

$$\rightarrow$$

$$\leftarrow$$

Final output:

Concatenation

Used in:

NLP

Speech

16. LSTM Variants

GRU (Gated Recurrent Unit)

Simpler version

Only 2 gates:

Update gate

Reset gate

Less parameters

Faster training

17. Applications of LSTM

NLP

Machine Translation

Example:

English → Hindi

Speech Recognition

Voice assistants

Time Series

Stock prediction

Weather prediction

Video processing

Action recognition

Text generation

ChatGPT earlier versions used LSTM

18. Advantages of LSTM

Can learn long-term dependencies

Solves vanishing gradient

Handles sequence data

High accuracy

19. Disadvantages

Slow training

Complex

Large memory usage

Parallelization difficult

20. Comparison: RNN vs LSTM

Feature	RNN	LSTM
Memory	Short	Long
Vanishing Gradient	Yes	No
Accuracy	Lower	Higher
Complexity	Simple	Complex

21. Computational Complexity

Per time step:

$$O(h(h+d))$$

Total:

$$O(Th(h+d))$$

22. LSTM in Deep Learning Frameworks

PyTorch:

```
torch.nn.LSTM
```

TensorFlow:

```
tf.keras.layers.LSTM
```

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23. Hyperparameters

Important ones:

Hidden size

Number of layers

Learning rate

Sequence length

Batch size

Dropout

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24. Example: Stock Prediction

Input:

Stock prices

Output:

Future price

LSTM learns:

Patterns

Trends

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25. Visualization Intuition

Think of LSTM like:

Human brain memory

Forget irrelevant things

Remember important things

26. Real World Example

Sentence:

"I live in India. I speak fluent Hindi"

LSTM remembers:

India → Hindi

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27. Mathematical Summary

Core:

$$C_t = f_t C_{t-1} + i_t \tilde{C}_t$$

$$h_t = o_t \tanh(C_t)$$

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28. Training Challenges

Overfitting

Solution:

Dropout

Gradient explosion

Solution:

Gradient clipping

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29. Initialization

Hidden state:

$$h_0 = 0$$

Cell state:

$$C_0 = 0$$

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30. Why Transformers Replaced LSTM

Problems:

Slow

Sequential

Cannot parallelize

Transformers solve:

Parallelization

Long context better

31. Interview Questions

Common ones:

What is LSTM?

Difference between LSTM and GRU

Why forget gate exists

Explain cell state

Why LSTM better than RNN

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32. Short Summary

LSTM is a special RNN

Uses gates

Stores long-term memory

Prevents vanishing gradient

Used for sequence tasks

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33. Diagram Flow (Text Form)

```
Input xt
↓

Forget Gate

↓

Input Gate

↓

Candidate

↓

Update Cell State

↓

Output Gate

↓

Hidden State
```

34. Formula Cheat Sheet

MOST IMPORTANT:

$$f_t = \sigma(W_f[h_{t-1}, x_t])$$

$$i_t = \sigma(W_i[h_{t-1}, x_t])$$

$$\tilde{C}_t = \tanh(W_c[h_{t-1}, x_t])$$

$$C_t = f_t C_{t-1} + i_t \tilde{C}_t$$

$$h_t = o_t \tanh(C_t)$$